

Exp. 1 : Matrix operation in C

Aim

- To write a C program to perform basic matrix operation such as addition, subtraction, multiplication and transpose of a matrix using two-dimensional arrays.

Objectives:-

- 1) To understand concept of 2-D arrays in C.
- 2) To perform arithmetic operation on matrices.
- 3) To implement matrix addition and subtraction.
- 4) To implement matrix multiplication using nested loops.
- 5) To find transpose of a matrix.
- 6) To develop programming skills using arrays and loop.

Theory

- A matrix is a rectangular arrangement of elements organized in rows and columns. In C programming, matrices are represented using 2-D arrays. Matrix operations are widely used in computer graphics, image processing, machine learning, scientific computing and engineering applications.

1) Matrix addition:

Condition → both matrices must have same no of rows and columns.

$$\text{Formula} \rightarrow C[i][j] = A[i][j] + B[i][j]$$

2) Matrix subtraction:

Condition → both matrices must have same dimensions

$$\text{Formula} \rightarrow C[i][j] = A[i][j] - B[i][j]$$

3) Matrix Multiplications:

Condition $\rightarrow$ no of columns of first matrix must be equal to the no of rows of second matrix.

Formula $\rightarrow C[i][j] = \sum (A[i][k] \times B[k][j])$
 where $k = 0$ to $n-1$

4) Transpose of matrix:

Transpose of matrix is obtained by interchanging its rows and column.

Formula: $A^T [i][j] = A[j][i]$

* FA Q's :-

1] Explain matrix addition, subtraction, multiplication, and transpose with suitable examples and state the condition required for each operations.

Ans a) Addition: Add corresponding elements.

condition: Both matrices must have same order (rows and columns)

Formula: $C[i][j] = A[i][j] + B[i][j]$

b) Subtraction: Subtract corresponding elements.

condition: Both matrices must have same order (rows and columns)

Formula: $C[i][j] = A[i][j] - B[i][j]$

c) Multiplication: Multiply rows of the first matrix with columns of the second.

condition: columns of the first matrix

= Rows of the second matrix

Formula: $C[i][j] = \sum (A[i][k] \times B[k][j])$
 where $k = 0$ to $n-1$

- Transpose: Interchanging rows and columns, no special condition.

2] how are 2D arrays used to represent matrices in C?

Ans Matrices are stored in 2D arrays (int A[R][C]) elements are accessed using $A[i][j]$, and nested loops are used to read, display and perform operations.

3] Algorithm for matrix multiplication (why compatibility is needed).

Ans Algorithm for matrix multiplication

- 1) Read both matrices
- 2) Check if columns of the first matrix equal rows of the second.
- 3) Use three nested loops to multiply and add elements
- 4) Store the result in the third matrix
- 5) Display the result

4] ~~Print result~~ Application of matrix operation in real world domain

Ans Application of matrix operations

- Computer graphics
 - Image processing
- Machine learning
- Engineering calculations
- Scientific computing.

Q : Role of nested loop in matrix operations.

Ans : Nested loops are used to access rows and columns of matrices. They help perform addition, subtraction, multiplication, and transpose efficiently.

~~12/12/22~~